

Learned Latent Curves: A Closed-Loop Framework for Corpus Embedding and Recursive Skill Auditing

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August 4, 2026

Abstract. We describe a family of three composed techniques used to audit and prioritize improvement work across a corpus of engineering artifacts (software skills, self-documentation files): (1) a *learned latent curve*, which re-expands a single 1-D ordering coordinate t into a fixed-width D -dimensional embedding via a Fourier basis with learned frequencies; (2) *curve-guided recursive self-improvement* (curve-guided RSI), which uses sparse cells of the fitted curve as a prioritization lens for gap-mapping and bounded fixpoint-style editing cycles; and (3) a self-documentation variant of the same pipeline, retargeted at heterogeneous memory-file corpora via a per-corpus primitive basis and an explicit item-granularity rule. We give the governing equations for each stage, the empirical validation obtained on a corpus of software skills (N=62-211) and a self-documentation corpus (N=17-154), and note the technique's distinction from an unrelated prior work on weight-space reparameterization for neural network optimization (arXiv:2607.09967) that shares only surface vocabulary (“curve”, “curvature”) and no derivation.

1 Introduction

Many corpus-auditing problems reduce to the same shape: N items live in a high-dimensional feature space, and a practitioner wants (a) a single scalar ordering of the items and (b) a fixed-width representation indexed by that ordering, so that interpolation and sparse-region detection are well defined. Dimensionality reduction (collapsing features to t) and dimensionality *expansion* (re-expanding t into an embedding) are usually treated as separate problems. This paper documents a single closed-form round trip between them and its use as the fitting stage of a larger closed-loop audit pipeline.

2 The Learned Latent Curve

For output dimension $j = 1, \dots, D$ (canonically $D = 384$) and 1-D coordinate $t \in [0, 1]$, the model is

$$z_j(t) = a_{j,0} + \sum_{m=1}^k (a_{j,m} \sin(2\pi f_m t) + b_{j,m} \cos(2\pi f_m t))$$

with k shared learned frequencies $f_1, \dots, f_k$ and per-output coefficients $a_{j,m}, b_{j,m}$. Stacked over j , this is a curve $\gamma: \mathbb{R} \rightarrow \mathbb{R}^D$. Writing the design vector

$$\phi(t) = [1, \sin(2\pi f_1 t), \cos(2\pi f_1 t), \dots, \sin(2\pi f_k t), \cos(2\pi f_k t)]$$

the model is $\gamma(t) = C \cdot \phi(t)$ with coefficient matrix C , giving a parameter count of

$$\gamma(t) = C \phi(t), \quad C \in \mathbb{R}^{D \times (1+2k)}$$

$$P = k + D(1 + 2k)$$

At $D = 384, k = 8$: $P = 8 + 384 \times 17 = 6,536$ parameters, compared against $N \times D$ target scalars.

2.1 Closed-form coefficient solve

With the frequencies f held fixed, C has a closed-form ridge solution

$$C^* = Z^\top \Phi (\Phi^\top \Phi + \lambda I)^{-1}$$

where Φ is an $N \times (1+2k)$ design matrix and Z is the $N \times D$ target matrix. Only the k frequencies require non-convex search; the closed-form ridge fit at any candidate frequency vector is a sanity floor for the gradient-descent fit.

2.2 Frequency parameterization

Frequencies are stored in an unconstrained form and mapped through a softplus to guarantee strict positivity and finite gradients near zero:

$$f_m = \text{softplus}(\tilde{f}_m) = \log(1 + e^{\tilde{f}_m}) > 0$$

2.3 Losses

Let $\mathbf{t}$ be the N training coordinates, $\mathbf{Z}$ the targets, and $\hat{\mathbf{Z}} = \gamma(\mathbf{t})$ the fitted curve evaluated at the training coordinates. The total objective is

$$\mathcal{L} = \mathcal{L}_{rec} + \mathcal{L}_{freq} + \mathcal{L}_{smooth} (+ \mathcal{L}_{orth})$$

$$\mathcal{L}_{rec} = \text{mean}((\hat{\mathbf{Z}} - \mathbf{Z})^2)$$

$$\mathcal{L}_{freq} = \lambda_f \cdot \text{mean}(f^2)$$

$$\mathcal{L}_{smooth} = \lambda_s \cdot \text{mean}((\gamma_{i+1} - 2\gamma_i + \gamma_{i-1})^2) \text{ on a dense grid}$$

with $\lambda_f \approx 10^{-4}$, $\lambda_s \approx 10^{-3}$. The smoothness term regularizes the curve *between* observed points, where reconstruction loss alone is blind.

2.4 Obtaining the coordinate $\mathbf{t}$

The default pipeline sets $\mathbf{t}$ from the top principal component of a z-scored feature matrix, mapped to $[0,1]$; a rank-uniformized variant replaces raw PC1 scores by their ranks to improve design-matrix conditioning. The explained-variance ratio of PC1 (or PC1+PC2 for a 2-D surface variant) is recorded as an explicit go/no-go gate, with $\text{PC1} \geq 0.40$ treated as evidence of usable low-rank structure.

3 Curve-Guided Recursive Self-Improvement

Curve-guided RSI composes the fitted curve with a gap-mapping and editing loop. Coordinates are reduced to a 2-D surface $(u,v) \in [0,1]^2$ via PC1+PC2 of a binary primitive-coverage matrix $\mathbf{C}$ (an $N \times 9$ binary matrix) lifted to $D = 384$ by a fixed seeded orthonormal projection $\mathbf{Q}$:

$$\mathbf{Z} = \mathbf{C} \mathbf{Q}^\top, \quad (u_i, v_i) = \text{PCA}_{1,2}(\mathbf{Z})_i$$

The $[0,1]^2$ plane is discretized into a 0.05×0.05 grid ($21 \times 21 = 441$ cells). For a cell $c = (u,v)$,

$$\text{neighbors}(c) := \{ i \in \text{corpus} : \|(u_i, v_i) - c\|_\infty \leq r \}$$

$$\text{is_sparse}(c) := |\text{neighbors}(c)| = 0$$

with default radius $r = 0.05$. Cells with zero neighbors are gap candidates; for each, a fresh-context gap-mapping subagent (focused on that single item, never the whole corpus) is dispatched, and any flagged “Extend” gap is closed via a bounded editing loop capped at three cycles per item. After all cycles, Stage 1 is re-run and the closed-loop claim is verified by comparing the sparse-cell count before and after:

$$\Delta_{sparse} = |\text{sparse_cells}|_{post} - |\text{sparse_cells}|_{pre} < 0$$

$\Rightarrow$ gaps closed.

4 Self-Documentation Variant

The self-documentation offshoot retargets the equations above at heterogeneous memory-file corpora (SELF.md, SELF-CHANGELOG.md, and eight additional preference/rule files), under the granularity rule that *each version is one corpus item* (a changelog entry, a memory-file section, or a named row). Each corpus receives its own 9-D binary primitive basis rather than sharing one basis across structurally distinct document types, since a single shared basis was found to flatten the per-corpus signal. Near-constant columns (coverage > 0.90 or < 0.10) are dropped before the PCA lift.

5 Empirical Validation

Corpus / fit	N	PC1(+PC2)	Holdout R ²	Status
Skill corpus, v1 (raw-content target)	62	0.40 (1-D)	−0.155	fail
Skill corpus, v2 (primitive-coverage target)	213	0.40 (1-D)	+0.183	pass
Skill corpus, v3 (2-D surface)	211	0.4655	+0.4655	headline pass
Skill corpus, v4 (sentence-transformer target)	211	0.0955	−0.005 to +0.130	fail (wrong target)
Curve-guided RSI, cycle 5 (69 skills)	69	0.4615	+0.2244	pass
SELF-CHANGELOG.md, canonical	17	0.9164	+0.9917	pass
SELF.md, canonical	51	0.6952	+0.7041	pass
Combined memory-file corpus, v4	154	0.6479	+0.7877	pass

Table 1. Fit quality across corpora. PC1(+PC2) is the explained-variance gate (Section 2); Holdout R² is measured against held-out items refit from their t coordinate alone.

6 Relation to Prior Work

The model in Section 2 shares no derivation with the weight-space reparameterization proposed in arXiv:2607.09967 (“Learning in Curved Weight Space: Exponential-Linear Weight Reparameterization for Improved Optimization”), which maps a single raw neural-network weight scalar w through a sign-aware symmetric-exponential/linear pathway pair to obtain an effective weight w_{eff} for optimizer training, and is validated by training-step reduction on autoregressive language models. The present work instead maps a per-item ordering coordinate t through a Fourier basis to obtain a fixed-width corpus embedding, and is validated by holdout reconstruction accuracy on document corpora. The overlap is limited to the word “curve”/“curvature” and the presence of learnable shape parameters; neither the domain, the basis function family, nor the fitting objective is shared. The two are unrelated by construction, not merely by citation omission — a full side-by-side comparison is documented separately [3].

Within the embedding literature, the model in Section 2 is a fitted generalization of two established fixed-basis constructions: random Fourier features [1] use *fixed random* frequencies as a kernel approximation, and transformer positional encodings [2] use *fixed geometric* frequencies with no learned coefficients. The present model instead learns both the frequencies and the coefficients from the target embeddings directly.

7 Conclusion

The learned latent curve gives a closed-form, auditable, interpolatable representation for small-N corpora with a low-rank ordering structure. Composed with sparse-cell detection and a bounded gap-closing loop, it converts a one-shot corpus audit into a measurable, repeatable improvement process with an explicit before/after metric. The same pipeline transfers, with corpus-specific primitive bases, from a software-skill corpus to a self-documentation corpus.

References

- [1] A. Rahimi and B. Recht. *Random Features for Large-Scale Kernel Machines*. Advances in Neural Information Processing Systems, 2007.
- [2] M. Tancik et al. *Fourier Features Let Networks Learn High Frequency Functions in Low Dimensional Domains*. Advances in Neural Information Processing Systems, 2020.
- [3] E. Smith. *Learning in Curved Weight Space: Exponential-Linear Weight Reparameterization for Improved Optimization*. arXiv:2607.09967 [cs.LG], 2026.